

Week 4: Evaluation Metrics and Performance Monitoring

Course: BUSM2725 AI and Analytics for Business Applications

Outline

- Overview of performance metrics.
- Activity: Measure the performance of ML models using house price and customer churn models
- Activity: Build a simple app using Gradio and Gemini to demonstrate how the model works and build a simple dashboard for monitoring performance,
- Q&A for Assessment 1.

Why Do We Need Metrics?

- Quantifying model quality
- Ensuring reliability, and
- Translating mathematical outputs into business value.



Why are metrics necessary?

They provide objective assessment, enable model comparison, and track improvement.



Made with  Napkin

What metrics can help us with?



Objective Assessment
Standardized measurement of learning



Model Comparison
Fair evaluation of different algorithms



Improvement Tracking
Monitoring progress during optimization

The Google Map team build an ML model to predict the duration of a trip:

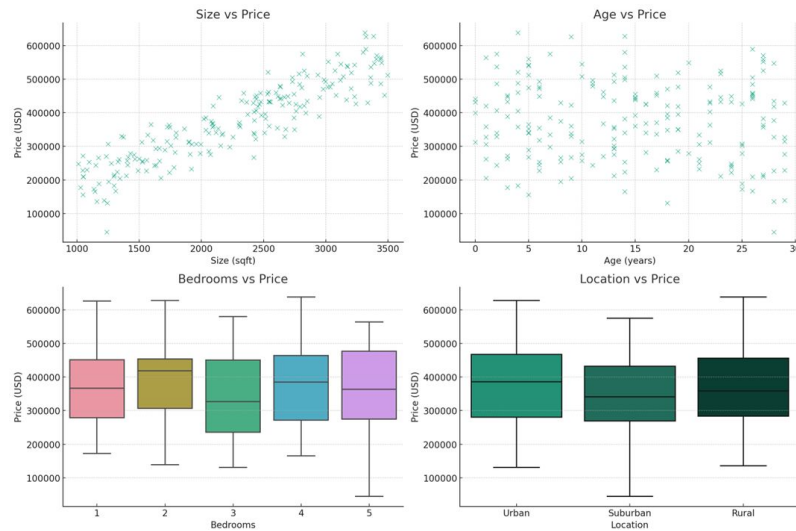
- What metrics should we use?
- Can the metrics help the team decide which model to use?
- Can the metrics help the team tracking the performance of the model during training and testing?

Let's re-examine two common ML tasks

	Classification	Regression
Goal	Predict discrete labels or categories	Predict continuous numerical values
Examples	Spam vs. Not Spam, Fraud Detection, Image Recognition	House Prices, Stock Value Prediction, Temperature Forecasting
How do we know if predictions are correct (or good)?		
In a sentence, define what metric is needed for this task.		

Regression Scenario: House price prediction (1)

- **Size (in square feet):** The total area of the house.
- **Number of Bedrooms:** Total bedrooms in the house.
- **Age of the House (in years):** How old the house is.
- **Location Type:** Categorizes the location as 'Urban', 'Suburban', or 'Rural'.
- **Price (in USD):** The price of the house.



See more details on Canvas (Before Class)

Regression Scenario: House price prediction (2)

Linear Regression model

$$\text{Price} = \beta_0 + \beta_1 \times \text{Size} + \beta_2 \times \text{Bedrooms} + \beta_3 \times \text{Age} + \beta_4 \times \text{Location}_1 + \beta_5 \times \text{Location}_2 + \dots + \epsilon$$

Here:

- β_0 is the y-intercept (the model's prediction when all features are 0).
- $\beta_1, \beta_2, \dots$ are the coefficients for each feature, representing the impact of each feature on the house price.
- $\text{Location}_1, \text{Location}_2, \dots$ are the one-hot encoded variables for the location.
- ϵ is the error term.



$$\begin{aligned} \text{Price (USD)} = & 48572.77 + 3803.02 \times \text{Location_Rural} - 13299.58 \times \\ & \text{Location_Suburban} + 9496.56 \times \text{Location_Urban} + 152.92 \times \\ & \text{Size (sqft)} - 4348.76 \times \text{Bedrooms} - 595.79 \times \text{Age (years)} \end{aligned}$$



Regression metrics

1. Mean Squared Error (MSE):

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

1. Mean Absolute Error (MAE):

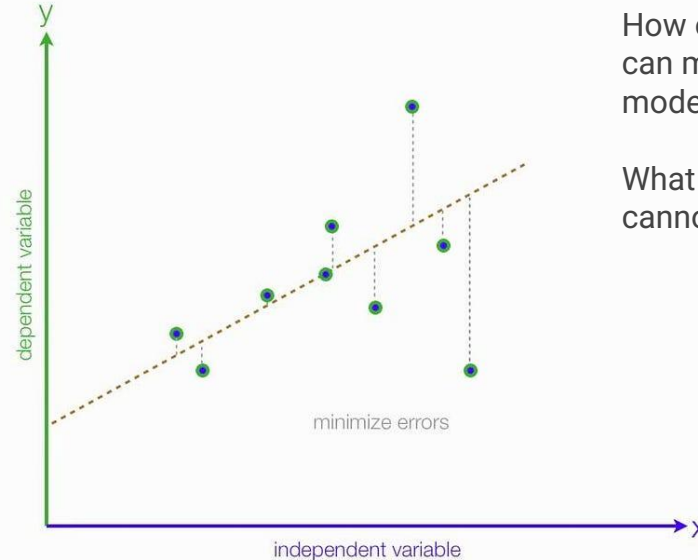
$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

1. Mean Absolute Percentage Error (MAPE):

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100\%$$

1. R-squared (R^2):

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$



How optimise each metric can make the prediction model better?

What the metrics can and cannot tell us?

Try Regression Game on Canvas

Model Parameters

Adjust coefficients to fit the line

Base Price (Intercept: β_0)

30000

Price per SqFt (Size: β_1)

105

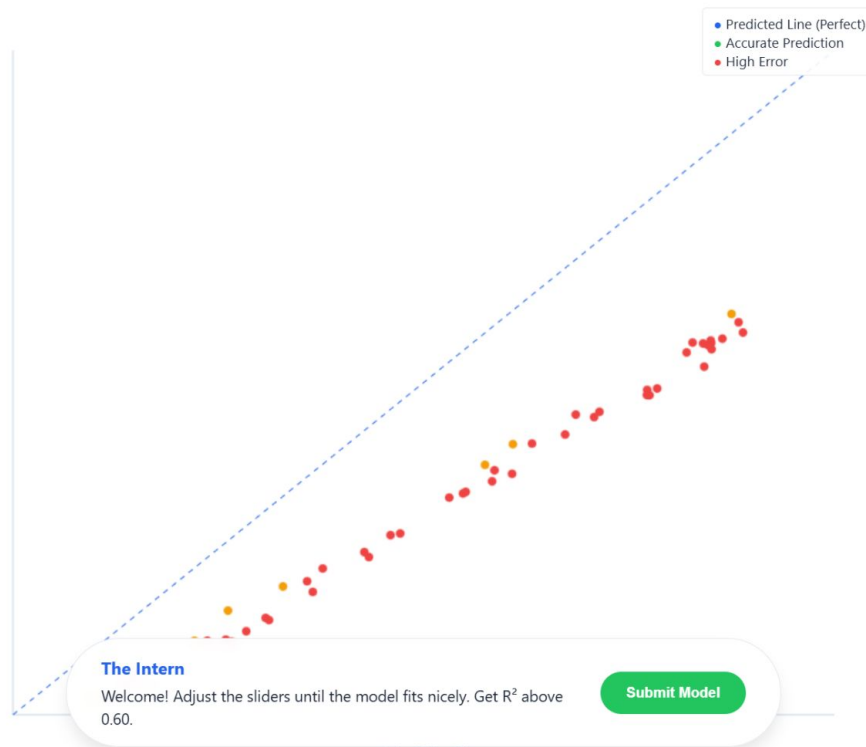
Price per Bedroom (β_2)

0

Price per Year Old (β_3)

-200

Predicted Price (\$)



Performance Metrics

Real-time Feedback

MAE (MEAN ABS ERROR)

\$101,561

Avg dollar error per house

MSE (MEAN SQ ERROR)

11,323,599,563

Penalizes large errors heavily

MAPE (%)

27.68%

Avg percentage error

R^2 (R-SQUARED)

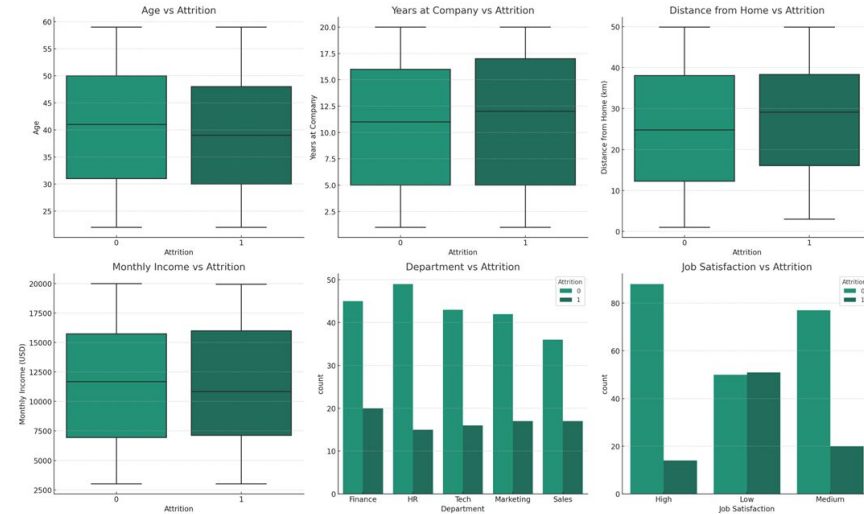
0.149

Goodness of fit (max 1.0)

Pro Tip: Look at the scatter plot. If points are above the line, increase coefficients. If below, decrease.

Classification Scenario: Predict employee attrition (1)

- **Age:** The age of the employee.
- **Department:** The department in which the employee works (e.g., Sales, HR, Tech).
- **Job Satisfaction:** A rating of the employee's job satisfaction (e.g., Low, Medium, High).
- **Years at Company:** How many years the employee has been with the company.
- **Distance from Home:** The distance from the employee's home to the workplace.
- **Monthly Income:** The employee's monthly income.
- **Attrition:** Whether the employee has left the company (Yes or No).



Classification Scenario: Predict employee attrition (2)

Decision tree

Rule 1: IF Job Satisfaction is Low (≤ 0.50) AND Distance from Home (km) ≤ 49.19 AND Monthly Income (USD) ≤ 3081.56 THEN Prediction: Left

Rule 2: IF Job Satisfaction is Low (≤ 0.50) AND Distance from Home (km) ≤ 49.19 AND Monthly Income (USD) > 3081.56 THEN Prediction: Stayed

Rule 3: IF Job Satisfaction is Low (≤ 0.50) AND Distance from Home (km) > 49.19 AND Distance from Home (km) ≤ 49.90 THEN Prediction: Left

...

Classification metrics

Accuracy:

Accuracy = (True Positives + True Negatives) / (True Positives + True Negatives + False Positives + False Negatives)

Precision:

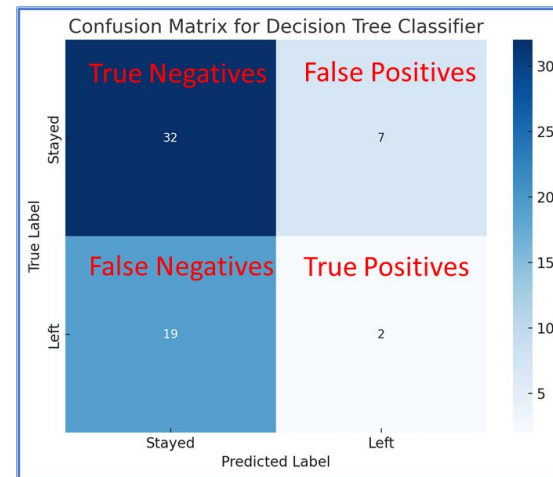
Precision = True Positives / (True Positives + False Positives)

Recall (Sensitivity or True Positive Rate):

Recall = True Positives / (True Positives + False Negatives)

F1-Score:

F1-Score = $2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$



Try Classification Game on Canvas


Metric Tycoon: HR Analytics
Level 1/4

Model Logic

1. CLASSIFICATION THRESHOLD

At what probability do we flag them as "Leaving"?

Loose Strict

Cutoff: 0.50

2. FEATURE IMPORTANCE (WEIGHTS)

Does this factor make people leave?

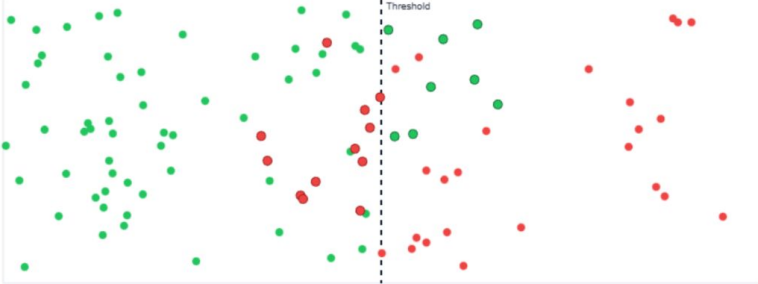
Distance from Home 10

Job Satisfaction (Low) 10

Low Income 10

Hint: Increase weights to separate the red/green dots. Adjust threshold to shift the vertical line.

Model Separation (Risk Scores)



Calculated Risk Score (0 to 1)

Confusion Matrix

	Pred: Stayed	Pred: Left
Actual: Stayed	56 True Negative	8 False Positive
Actual: Left	The First Day Welcome to HR. Just get the model **Accuracy**.	Submit Analysis

Metrics

SCORECARD

ACCURACY
80%
Overall correctness

PRECISION
75%
Quality of 'Left' flags

RECALL
67%
Quantity of 'Left' found

F1 SCORE
0.71
Balance of Prec & Rec

Activity: Build a Simple App using Gradio and Gemini

- Use Telco-Customer-Churn.csv data that we explore last week
- Goal: Use Gemini and Google Colab to help you implement an end-to-end data science project from (1) data understanding (2) model building (3) testing and deployment.
- Expected outcomes:
 - Help you gain high-level understanding of a data science process
 - Understand the choice of ML models and performance metrics
 - Understand how to work with AI to automate data science process
 - Appreciate the strengths and limitations of AI in data science process
 - Gain abilities to interact with AI to enhance output quality

Steps

- Get the Telco-Customer-Churn.csv file and upload it to Google Colab (review last week slides if you cannot remember how to do it)
- Use the following prompt with the AI coding agent in Google Colab:

You are a professional data scientist. Help me load the Telco-Customer-Churn.csv file, examine what columns are included. The goal is to develop machine learning models to predict Churn. Perform the following high-level steps: (1) data understanding (2) model building (build at least 2 interpretable ML models such as logistics regression or decision trees and one complex one like gradient boosting) (2) deploy the models for internal testing using Gradio.

 You are a professional data scientist. Help me load the Telco-Customer-Churn.csv file, examine what columns are included. The goal is to develop machine learning models to predict Churn. Perform the following high-level steps: (1) data understanding (2) model building (build at least 2 interpretable ML models such as logistics regression or decision trees and one complex one like gradient boosting) (2) deploy the models for internal testing using Gradio.

◆ Great! I've outlined a comprehensive plan to tackle your Telco customer churn prediction project, from data understanding to model deployment. Take a look and let me know if any adjustments are needed.

- **Load and Examine Data:** Load the 'Telco-Customer-Churn.csv' file into a pandas DataFrame. Display the first few rows, column names, data types, and check for missing values to understand the dataset's structure and quality.
- **Preprocess Data for Modeling:** Perform necessary data preprocessing steps, including handling categorical variables (e.g., one-hot encoding), scaling numerical features if required, and addressing any missing values. This will prepare the data for training machine learning models.
- **Split Data:** Divide the preprocessed dataset into training and testing sets to ensure robust model evaluation.
- **Train Interpretable Models:** Train at least two interpretable machine learning models, such as Logistic Regression and Decision Tree Classifier, on the training data to predict 'Churn'.
- **Train Complex Model:** Train a complex machine learning model, such as Gradient Boosting Classifier, on the training data to predict 'Churn'.
- **Evaluate Model Performance:** Evaluate the performance of all trained models using appropriate metrics (e.g., accuracy, precision, recall, F1-score, ROC-AUC) on the test set. Provide a comparison of the model performances.
- **Deploy Models with Gradio:** Develop a Gradio interface to deploy the trained models, allowing for interactive testing and prediction. This interface will be used for internal testing purposes.
- **Final Task:** Summarize the findings from the data analysis, the performance of the developed models, and the insights gained from the project.

Generating...

↳ Preparing

- ↳ Load and Examine Data
- ↳ Preprocess Data for Modelling
- ↳ Split Data
- ↳ Train Interpretable Models
- ↳ Train Complex Model
- ↳ Evaluate Model Performance
- ↳ Deploy Models with Gradio
- ↳ Final Task

Example outputs

- Notes: AI generated plan (content) will not be the same from your end
- Below are steps that AI agent want to execute (I allowed auto-run mode)
 - **Load and Examine Data:** Load the 'Telco-Customer-Churn.csv' file into a pandas DataFrame. Display the first few rows, column names, data types, and check for missing values to understand the dataset's structure and quality.
 - **Preprocess Data for Modeling:** Perform necessary data preprocessing steps, including handling categorical variables (e.g., one-hot encoding), scaling numerical features if required, and addressing any missing values. This will prepare the data for training machine learning models.
 - **Split Data:** Divide the preprocessed dataset into training and testing sets to ensure robust model evaluation.
 - **Train Interpretable Models:** Train at least two interpretable machine learning models, such as Logistic Regression and Decision Tree Classifier, on the training data to predict 'Churn'.
 - **Train Complex Model:** Train a complex machine learning model, such as Gradient Boosting Classifier, on the training data to predict 'Churn'.
 - **Evaluate Model Performance:** Evaluate the performance of all trained models using appropriate metrics (e.g., accuracy, precision, recall, F1-score, ROC-AUC) on the test set. Provide a comparison of the model performances.
 - **Deploy Models with Gradio:** Develop a Gradio interface to deploy the trained models, allowing for interactive testing and prediction. This interface will be used for internal testing purposes.
 - **Final Task:** Summarize the findings from the data analysis, the performance of the developed models, and the insights gained from the project.

Pause and review the steps

- Are you happy with the plan?
- Do you think that the plan is clear enough or any steps that might benefit from more detailed instructions?
- Do you think that you will get the outputs you want?
- Revise the plan if needed

Gradio App

Telco Customer Churn Predictor

Enter customer details to predict churn likelihood.

Gender
Female

Senior Citizen
0: No, 1: Yes
☐ 0 ☐ 1

Partner
No

Dependents
No

Tenure (months)
1 72

Phone Service
No

Multiple Lines
No phone service

Internet Service
DSL

Monthly Charges
50

Total Charges
1000

Clear Submit

output
Customer is predicted NOT to churn with a probability of 0.55

Flag

What is app is used for?
Is it helpful for internal testing?
Which ML model is used for
power the app?

How to improve the app?

Improved Testing

Single Customer Prediction

Batch Evaluation

Telco Customer Churn Batch Evaluator

Upload a CSV file to evaluate the churn prediction model and see performance metrics.

Upload CSV for Batch Prediction

Telco-Customer-Churn-Testing.csv

13.8 KB

Clear

Submit

Accuracy

Accuracy: 0.8283

Precision

Precision: 0.6842

Recall

Recall: 0.5417

F1-Score

F1-Score: 0.6047

ROC-AUC

ROC-AUC: 0.8850

Flag

Follow-up prompt:

Improve the gradio app by (1) using sliders for numeric variables (2) allow user to upload testing dataset and visualise the performance metrics with testing data

Q&A for Assessment 1

- Please carefully document the prompts you used for analyses (clarify your assumptions and expected outcomes)
- Explain in the report how to verify the outcomes generated by AI
- You can use the outputs from Colab to conduct further analysis on your own rather relying entirely on AI (this is desirable for complex tasks)